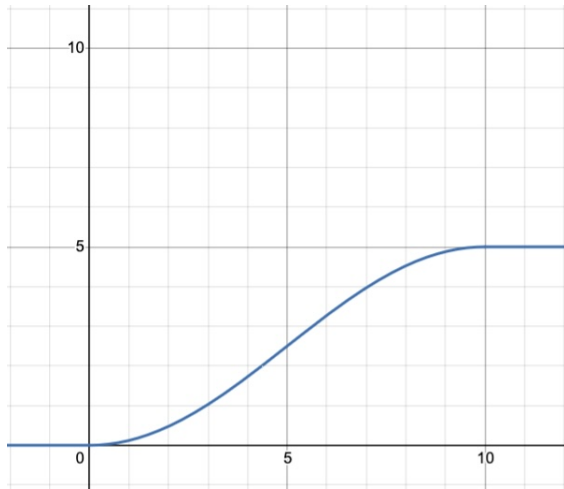


Learning target 1.1, version 1

Compute and interpret average velocity and its units, using information about the position function (as a table, graph, and/or function formula).

The position (in miles), $s(t)$, of a car driving along a straight road at time t (in minutes), is given by the following graph.



1. Determine the average velocity of the car between $t = 5$ and $t = 10$ minutes, $AV_{[5,10]}$. Use proper notation and the work you did to determine the result; include units on your answer.
2. On the graph of $s(t)$ draw a line through $(3, s(3))$ and $(7, s(7))$; what is the slope of this line and what does that slope mean in the physical context of the function s ?
3. Here's some additional data for the function $s(t)$ that's pictured above:

t (in minutes)	8.0	8.05	8.1	8.15
$s(t)$ (in miles)	4.52254	4.54537	4.56770	4.58952

Find the average velocity of the car on the interval $[8.05, 8.1]$. Label your result using proper notation and include units on your answer.

Learning target 1.1, version 2

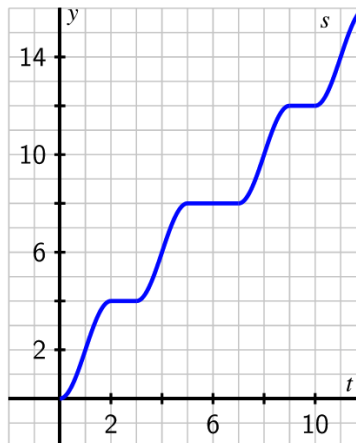
Compute and interpret average velocity and its units, using information about the position function (as a table, graph, and/or function formula).

1. Your fitbit records some data about the distance you traveled on a bike ride around the city:

t (in minutes)	0	15	30	45	60	75
$d(t)$ (in miles)	0	4.5	10.5	15	21	24

Find your average velocity over the time interval from $t = 30$ to $t = 75$. Include units.

2. The following graph shows a plot of the function $s(t)$ that measures the location of a car (in thousands of feet) along a road at time t in minutes.



- Use the graph to find the average velocity of the car between $t = 2$ and $t = 7$ minutes, $AV_{[2,7]}$. Include units on your answer.
- On the graph of $s(t)$ draw a line through $(7, s(7))$ and $(10, s(10))$; what is the slope of this line, and what does that slope mean in the physical context of the function s ? Include units on your answer.